

对 M 进行分析:

$$\text{沿斜面向上: } \sin\theta T + \cos\theta N = mg$$

$$\text{沿斜面向下: } \cos\theta T = ma + \sin\theta N$$

$$\cos^2\theta ma$$

$$\cos\theta T = ma + mg \sin\theta \cos\theta - ma \sin^2\theta$$

$$ma \sin\theta + N = mg \cos\theta \Rightarrow N = m(g \cos\theta - a \sin\theta)$$

$$T = \cos\theta ma + mg \sin\theta = m(a \cos\theta + g \sin\theta)$$

(2) 当小球离开斜面, 有 $N=0 \Rightarrow g \cos\theta = a \sin\theta \Rightarrow a = \frac{g}{\tan\theta}$

2. (1) 分析 m: $a_m = \frac{N \sin\theta}{m}$

分析 M: $a_M = \frac{N \sin\theta}{M}$

有: $a_M : a_m = m : M$ 故经过时间 t: $V_M : V_m = m : M$

同理: $X_M : X_m = m : M$ 有 $X_M + X_m = \frac{h}{\tan\theta}$

$$\therefore X_M = \frac{m}{m+M} \cdot \frac{h}{\tan\theta} = \frac{-mh}{(m+M)\tan\theta} \text{ (向左)}$$

$$a_1 \left(\frac{M}{\sin\theta} + m \sin\theta \right) = mg \cos\theta$$

$$\frac{Ma_1}{\sin\theta} + ma_1 \sin\theta = mg \cos\theta$$

(2) 以水平向右为正方向, 向下为正方向

对 M: $Ma_1 = N \sin\theta \Rightarrow N = \frac{Ma_1}{\sin\theta} \quad ①$

对 m: 垂直于斜面方向上, 有: $F_{\perp} = 0 \Rightarrow N + ma_1 \sin\theta = mg \cos\theta \quad ②$

$$mg \sin\theta + ma_1 \cos\theta = ma_2 \quad ③$$

联立 ①②③. $a_1 = \frac{mg \cos\theta}{\frac{M}{\sin\theta} + m \sin\theta} = \frac{mg \sin\theta \cos\theta}{M + m \sin^2\theta}$

(3) 以 M 作为参考系: 对 m: 有
$$\begin{cases} N + ma_1 \sin\theta = mg \cos\theta \\ mg \sin\theta + ma_1 \cos\theta = ma_2 \end{cases} \Rightarrow$$

$$a_2 = g \sin\theta + a_1 \cos\theta = g \sin\theta + \frac{mg \sin\theta \cos^2\theta}{M + m \sin^2\theta}$$

$$= g \sin \theta \left(1 + \frac{m \cos^2 \theta}{M + m \sin^2 \theta} \right) = g \sin \theta \cdot \left(\frac{M + m \sin^2 \theta + m \cos^2 \theta}{M + m \sin^2 \theta} \right) = \frac{(M+m)g \sin \theta}{M + m \sin^2 \theta}$$

$$14) \text{ 对 } m: a'_{2x} = \cos \theta a'_2 = \frac{(M+m)g \sin \theta \cos \theta}{M + m \sin^2 \theta}$$

$$a'_{2y} = \sin \theta a'_2 = \frac{(M+m)g \sin^2 \theta}{M + m \sin^2 \theta}$$

$$a'_{2y} = a_{2y}$$

$$a_{2x} = a'_{2x} - a_1 = \frac{(M+m)g \sin \theta \cos \theta + mg \sin \theta \cos \theta}{M + m \sin^2 \theta} = \frac{Mg \sin \theta \cos \theta}{M + m \sin^2 \theta}$$

$$a_2 = \sqrt{a_{2x}^2 + a_{2y}^2} = \frac{\sqrt{(M+m)^2 g^2 \sin^4 \theta + M^2 g^2 \sin^2 \theta \cos^2 \theta}}{M + m \sin^2 \theta}$$

方向与地成 $\arctan \frac{M \cos \theta}{(M+m) \sin \theta}$

$$15) N = \frac{M a_1}{\sin \theta} = \frac{M m g \sin \theta \cos \theta}{M + m \sin^2 \theta} \cdot \frac{1}{\sin \theta} = \frac{M m g \cos \theta}{M + m \sin^2 \theta}$$

$$16) R = Mg + \cos \theta N = Mg + \frac{M m g \cos^2 \theta}{M + m \sin^2 \theta} = \frac{M^2 g + M m g}{M + m \sin^2 \theta}$$

$$3. \text{ 从上往下, } f_0 = 0.3 \times 3 \times 10^{-3} \times 9.8 =$$

$$a_1 = 1$$

$$d = \psi$$

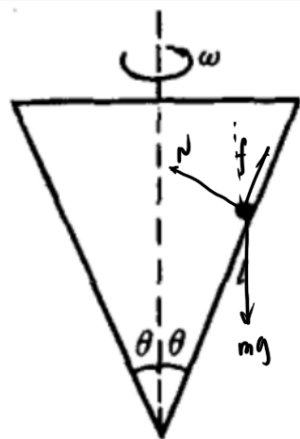
$$a_{299} = 1 + 4 \times 298$$

只考虑左边的书. 第一页: $f_1 = f_0$. 第二页: $f_2 = 2 \times f_0 + 3 \times f_0 = 5f_0$

第三页: $f_3 = 4f_0 + 5f_0 = 9f_0 \dots$ 第299页: $f_{299} = 1193f_0$. 第300页: $f_{300} = 1197f_0$

$$\therefore F = f_1 + \dots + f_{300} = f_0 + 5f_0 + \dots + 1197f_0 = 179700 f_0$$

$$\therefore F = 1584.954$$

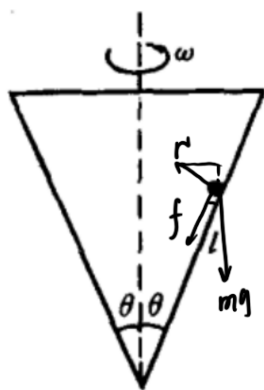


1. 若摩擦力向上: $f = \mu N$. $a_n = \omega^2 R$. $R = l \cos \theta$

$$\cos \theta f + \sin \theta N = mg \Rightarrow \cos \theta \mu N + \sin \theta N = mg \Rightarrow N = \frac{mg}{\mu \cos \theta + \sin \theta}$$

$$\cos \theta N - \sin \theta f = \mu a_n = \cos \theta N - \mu \sin \theta N = N(\cos \theta - \mu \sin \theta) = \frac{mg}{\mu \cos \theta + \sin \theta} (\cos \theta - \mu \sin \theta)$$

$$\therefore \omega^2 l \cos \theta = \frac{g(\cos \theta - \mu \sin \theta)}{\mu \cos \theta + \sin \theta} \Rightarrow l = \frac{g(\cos \theta - \mu \sin \theta)}{(\mu \cos \theta + \sin \theta) \cos \theta \omega^2}$$



2° 若摩擦力向下: $f = \mu N$. $a_n = \omega^2 l \cos \theta$

$$\sin \theta N = \cos \theta f + mg = \mu \cos \theta N + mg \Rightarrow N = \frac{mg}{\sin \theta - \mu \cos \theta}$$

$$\omega^2 l \cos \theta = \cos \theta N + \sin \theta f = \cos \theta N + \mu \sin \theta N = (\cos \theta + \mu \sin \theta) N = (\cos \theta + \mu \sin \theta) \frac{mg}{\sin \theta - \mu \cos \theta}$$

$$\therefore l = \frac{g(\cos \theta + \mu \sin \theta)}{(\sin \theta - \mu \cos \theta) \cos \theta \omega^2}$$

$$\therefore \frac{g(\cos \theta - \mu \sin \theta)}{(\mu \cos \theta + \sin \theta) \cos \theta \omega^2} \leq l \leq \frac{g(\cos \theta + \mu \sin \theta)}{(\sin \theta - \mu \cos \theta) \cos \theta \omega^2}$$

5. 对于电梯中的人: $F_B = mg + \frac{1}{2}mg = \frac{3}{2}mg$ $\therefore a_B = \frac{\frac{3}{2}mg - T}{m} = \frac{3}{2}g - \frac{T}{m}$

$F_A = T$ $\therefore a_A = \frac{T}{m} = a_B$ $\therefore \frac{2T}{m} = \frac{3}{2}g \Rightarrow a_A = a_B = \frac{3}{4}g$

对电梯外的人: $a_{地B} = a_{电梯} + a_{梯B} = -\frac{1}{2}g + \frac{3}{4}g = \frac{1}{4}g$

$\therefore a_B = \frac{1}{4}g$ $F_B = \frac{1}{4}mg = mg - T \Rightarrow T = \frac{3}{4}mg$

$\therefore a_A = \frac{3}{4}g$

$\therefore a_A = \sqrt{(\frac{3}{4}g)^2 + (\frac{1}{2}g)^2} = \sqrt{\frac{9}{16}g^2 + \frac{4}{16}g^2} = \frac{\sqrt{13}}{4}g$

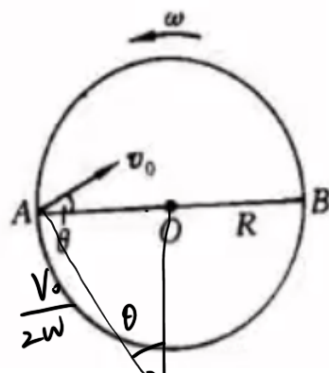
$\therefore$ 相对地: $\begin{cases} a_A = \frac{\sqrt{13}}{4}g \\ a_B = \frac{1}{4}g \end{cases}$ 相对梯: $\begin{cases} a_A = \frac{3}{4}g \\ a_B = \frac{1}{4}g \end{cases}$

6. 以圆岛为参考系. $F_{惯} = m\omega^2 R$ $F_{科} = -2m\vec{\omega} \times \vec{v}_0 = -2m|\vec{\omega}| \cdot |\vec{v}_0| = -2m\omega v_0$

又: $\omega R \ll v_0$. $\therefore |F_{科}| \gg F_{惯}$. 可忽略 $F_{惯}$.

又: $F_{科} = 2m\vec{\omega} \times \vec{v}_0$. 故有 $F_{科}$ 与 v_0 垂直并向下.

$\therefore$ 故圆周运动 $\frac{mv_0^2}{R} = F \Rightarrow \frac{mv_0^2}{R} = 2m\omega v_0$



$$\therefore \frac{V_0}{r} = 2\omega \Rightarrow r = \frac{V_0}{2\omega} \Rightarrow \sin\theta = \frac{\frac{R}{\frac{V_0}{2\omega}}}{\frac{V_0}{2\omega}} = \frac{2\omega R}{V_0} \Rightarrow \theta = \arcsin\left(\frac{2\omega R}{V_0}\right)$$

轨迹为一个圆. $r = \frac{V_0}{2\omega}$. 圆心在O正下方 $\sqrt{\frac{V_0^2}{4\omega^2} - R^2}$ 处